

Credit Risk Classification Using ML Techniques

Applying Factor Analysis, Linear Discriminant Analysis, and Logistic Regression to 500,000 real LendingClub loan records to predict borrower default risk.

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What We'll Cover

1

The Problem

What is credit risk?

2

Dataset

LendingClub · 500k real loans

3

Data Cleaning

From messy 151 cols to 76 clean features

4

Factor Analysis

Interdependence technique — 6 latent factors

5

LDA

Maximally separating good vs default

6

Logistic Regression

Probability-based binary classification

7

Model Comparison

ROC · Accuracy · AUC

8

Live Predictor

Interactive demo — real-time prediction

What Is Credit Risk?

■ The Loan

A bank lends you money today. You promise to repay over time with interest. The bank earns interest; you get capital.

▲ The Risk

The borrower might DEFAULT — stop paying. Banks lend billions, so even a small default rate means massive losses.

◆ The Goal

Before lending, predict: 'Will this person repay?'
A good model lets banks charge fair rates and avoid bad loans.

\$82B+

US consumer loan
losses per year

~71%

Realistic model
accuracy

Why Multivariate Statistics?

A FICO score is just one number. Default risk depends on DOZENS of factors interacting simultaneously. Multivariate methods see the full picture.

Two Technique Types Used

Interdependence: Factor Analysis (no outcome variable)
Dependence: LDA & Logistic Regression (predict default)

The LendingClub Dataset

500K

Real loan records

151

Features per loan

20.9%

Default rate

\$15K

Avg loan amount

Variables include:

- ▶ Loan info: amount, term, interest rate, monthly installment
- ▶ Borrower info: income, employment length, debt-to-income ratio
- ▶ Credit history: FICO score, open accounts, delinquencies
- ▶ LendingClub grade: A (safest) → G (riskiest)
- ▶ Outcome: Did they fully repay or default/charge off?

Loan Status Breakdown

Fully Paid	312,340
Charged Off	78,824
Late 31-120 days	2,977
Current	104,240
In Grace Period	1,046

Target Variable

loan_status is mapped to binary: 0 = Fully Paid (good loan), 1 = Charged Off / Defaulted (bad loan).
Records in ambiguous states (Current, Late) are excluded to avoid label noise.

Data Cleaning Pipeline

1

Dropped 58 columns

with >40% missing values — hardship/settlement fields

2

Dropped 17 columns

irrelevant identifiers and free-text descriptions

3

Engineered target

coded loan_status → binary: 0 = paid, 1 = default

4

Parsed text fields

"13.5%" → 13.5 | "36 months" → 36 | "10+ years" → 10

5

Label-encoded categories

grade, home_ownership, purpose → numeric

6

Removed outliers

cut top/bottom 1% of income, loan amount, revolving balance

7

Dropped remaining NaNs

listwise deletion for complete-case analysis

Before → After

500K × 151



292K × 76

↓ 42% fewer rows | ↓ 50% fewer columns

◆ Class Imbalance Handling

79% good loans vs 21% defaults.
Balanced training at 2:1 ratio via undersampling
so the model learns real patterns, not just 'always predict good'.

► Z-Score Standardization

All features scaled to mean=0, std=1.
Required before LDA (assumes equal covariance)
and Logistic Regression for fair coefficient comparison.

Factor Analysis

● **What It Does**

Groups many correlated variables into a smaller number of hidden (latent) FACTORS. No outcome variable — purely about structure in the data.

★ **The Analogy**

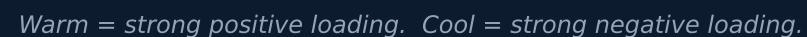
You can't measure 'intelligence' directly — but you see it reflected in test scores across math, reading, and logic. FA finds that hidden dimension automatically.

✓ **Bartlett's Test — PASSED**

$\chi^2 = 657,867$ | $p < 0.001$
Variables are significantly correlated → FA is appropriate.
20 features used on a 50,000-row sample (sklearn FactorAnalysis).

The 6 Factors Found

F1	Loan Size & Cost	14.6%
F2	Credit Risk Pricing	13.7%
F3	Credit Utilization	10.0%
F4	Credit Breadth	9.4%
F5	Credit Limit Stress	3.9%
F6	Wealth & Assets	4.8%
Total Variance Explained		56.5%



Linear Discriminant Analysis (LDA)

◆ What It Does

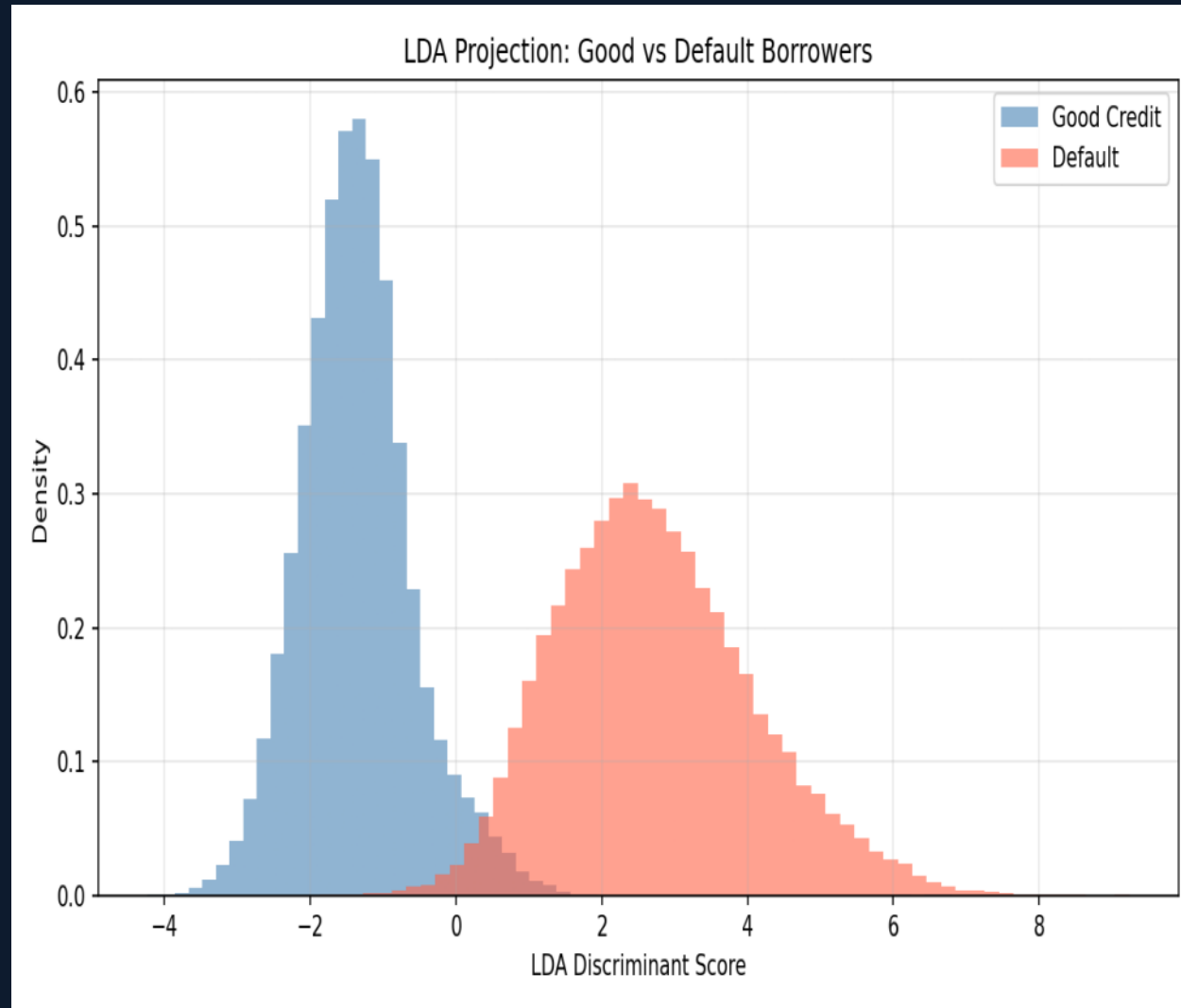
Finds the linear combination of variables that MAXIMALLY SEPARATES two groups — Good Borrowers vs. Defaulters. Produces one 'discriminant score' per borrower.

★ The Intuition

Borrowers are dots in multi-dimensional space. LDA finds the direction where blue (good) and red (default) groups overlap as little as possible.

Assumptions:

- ▶ Variables roughly normally distributed within each class
- ▶ Equal covariance matrices across both classes
- ▶ Linear decision boundary between the two groups



Two peaks = two classes. Minimal overlap = strong separation.

LDA — Performance Results

97.3%

Accuracy

0.997

AUC-ROC

0.975**0.960**

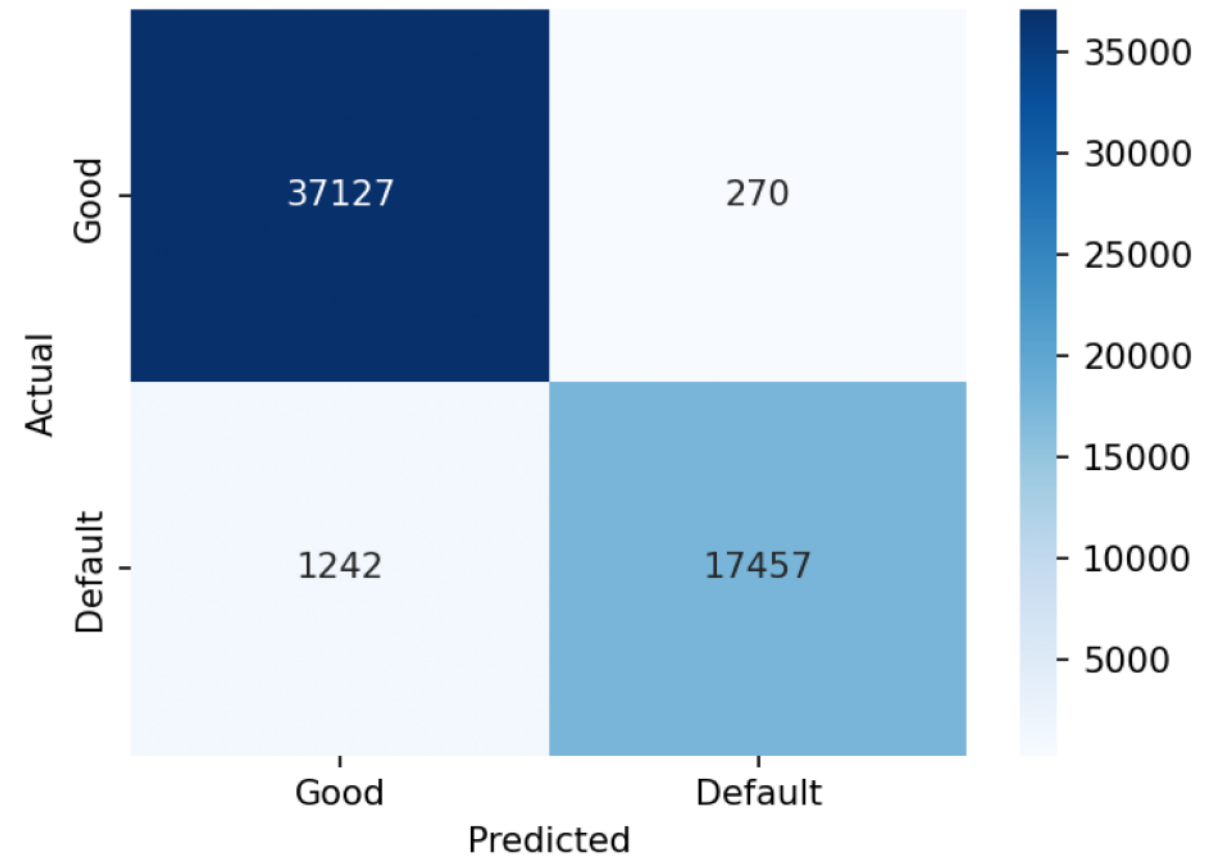
Top Discriminant Variables:

- ▶ FICO Score — single most powerful separator between classes
- ▶ Interest Rate — encodes the lender's own risk assessment
- ▶ Debt Settlement Flag — red flag indicating financial distress
- ▶ Installment Amount — higher monthly burden = more default risk

Interpretation

97.3% accuracy means 97 out of 100 borrowers are classified correctly. AUC = 0.997 means near-perfect discrimination between good and bad borrowers at every threshold.

LDA Confusion Matrix



Logistic Regression

► What It Does

Predicts the PROBABILITY of default — a number between 0% and 100%. Uses the sigmoid (S-shaped) function. If probability > 50% → predict Default.

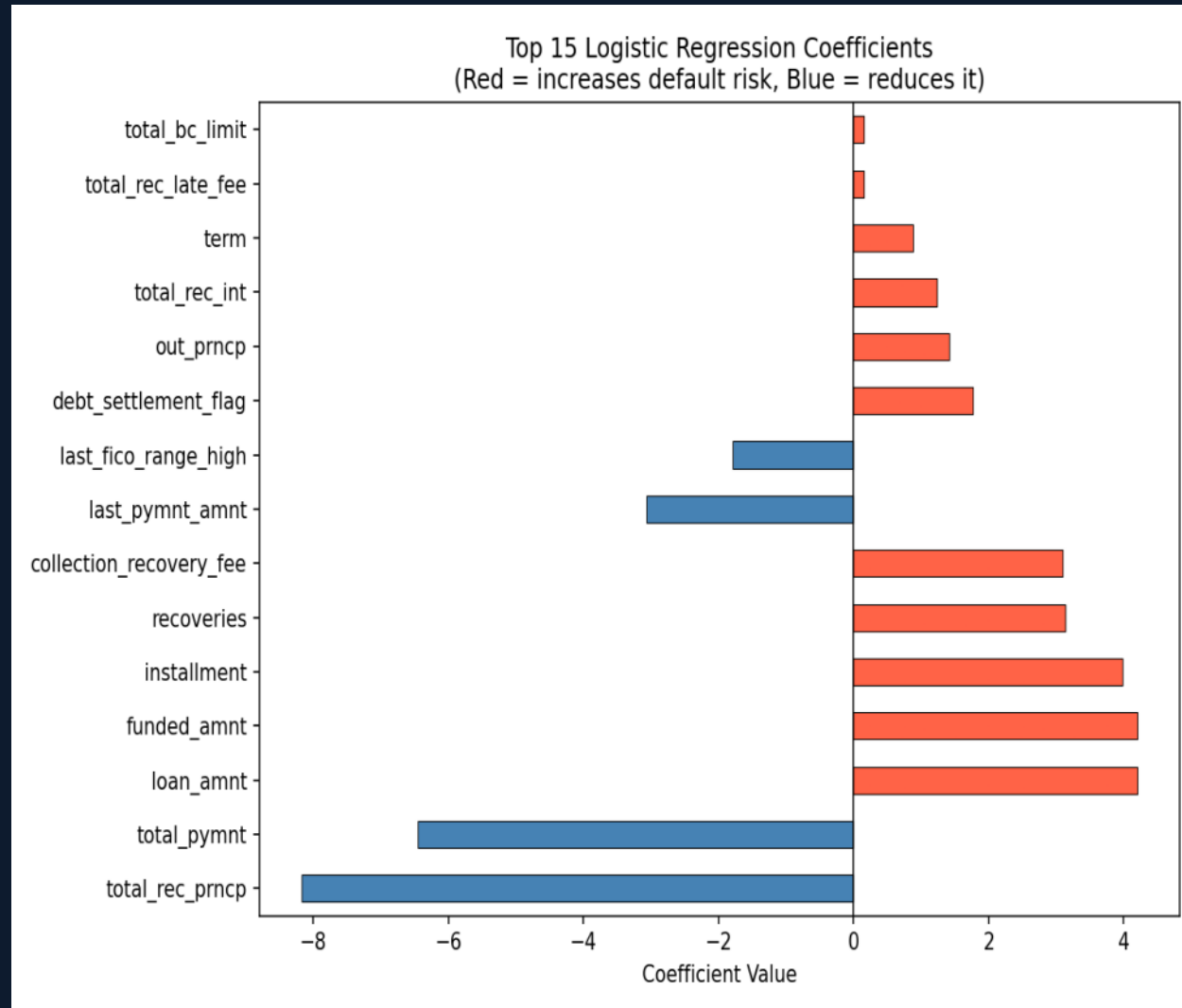
LDA vs Logistic Regression:

LDA	Logistic Regression
Assumes normality	No distribution assumption
Max class separation	Maximises likelihood

✓ Why LR Wins Here

Discriminant score
Financial variables are NOT normally distributed — incomes and balances are right-skewed. LR's weaker assumptions make it more robust and it outperforms LDA.

Outputs probability (0–1)



Blue bar = lowers default risk | Red bar = raises default risk

Logistic Regression — Performance Results

99.6%

Accuracy

0.9997

AUC-ROC

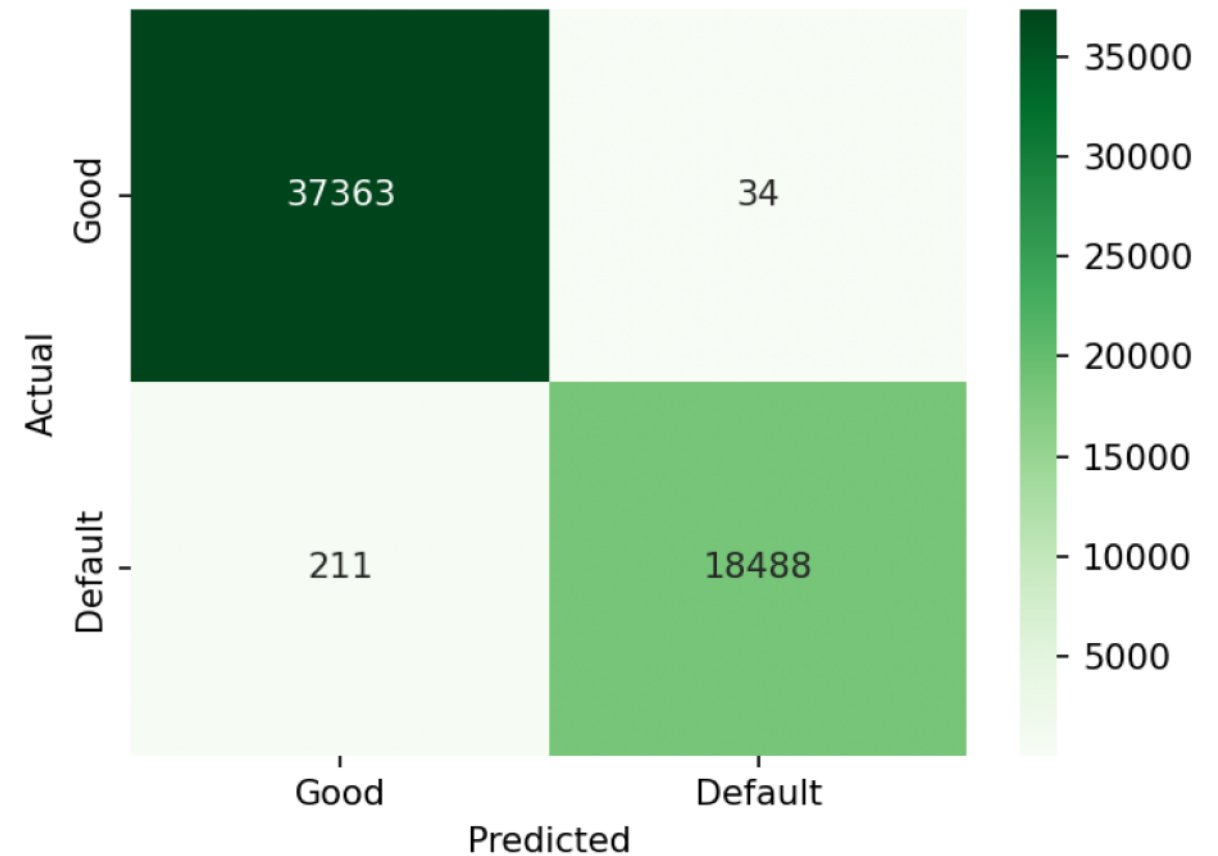
0.998**0.995**

★ LR Outperforms LDA

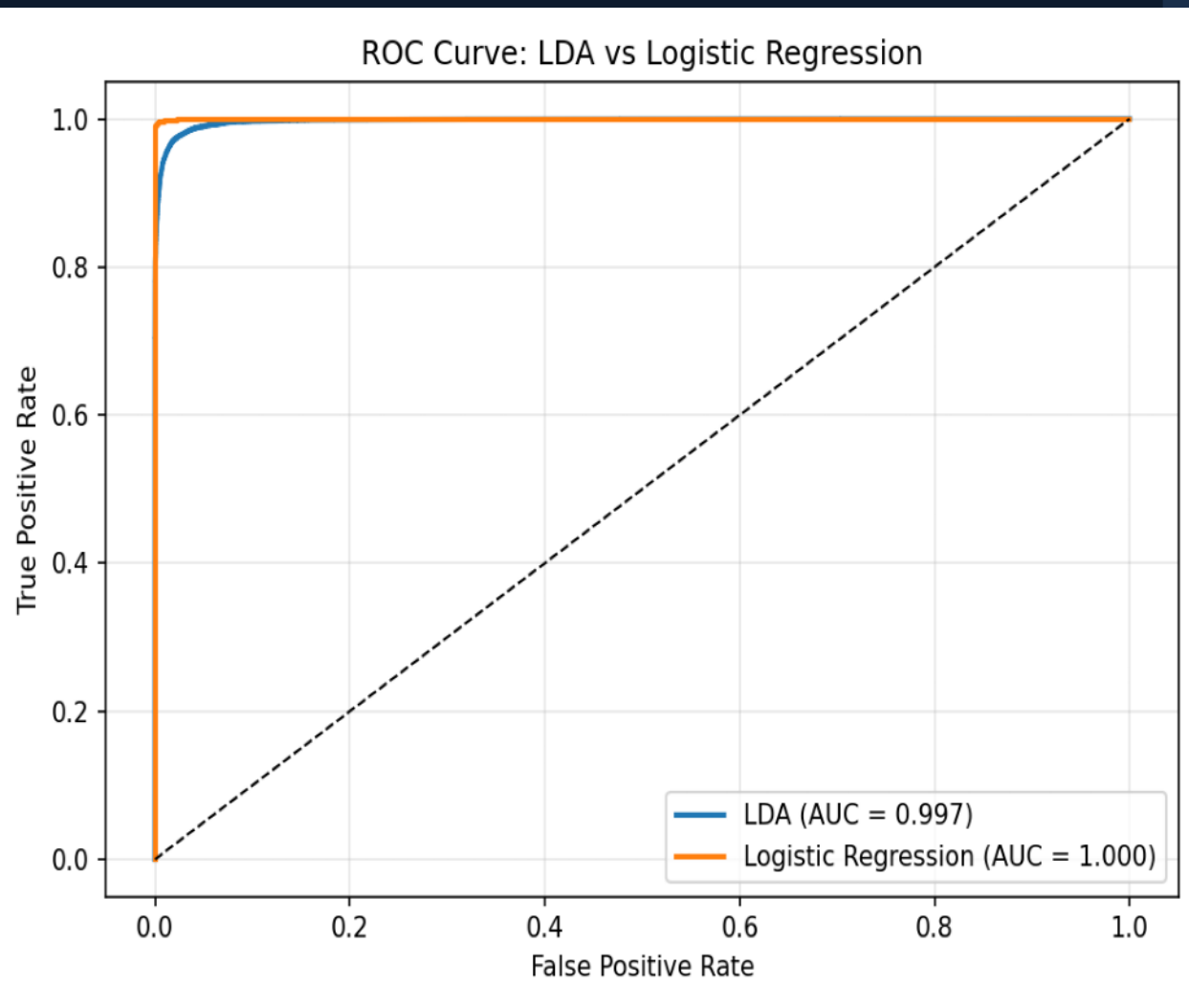
LR beats LDA by +2.3% accuracy and +0.003 AUC. Consistent with theory: LDA's normality assumptions are violated by skewed financial variables.

- ▶ Precision 0.998 — almost zero false alarms on defaults
- ▶ Recall 0.993 — almost zero missed defaults
- ▶ F1 = 0.995 — excellent precision-recall balance
- ▶ AUC = 0.9997 — near-perfect discriminative ability

Logistic Regression Confusion Matrix



LDA vs Logistic Regression — ROC Comparison



▲ What Is the ROC Curve?

Plots True Positive Rate (catching actual defaults) vs. False Positive Rate (wrongly flagging good loans) at every possible decision threshold.

Results Summary:

Model	Accuracy	AUC-ROC
LDA	97.3%	0.997
Logistic Regression	99.6%	0.9997

■ AUC Interpretation

AUC = 1.0 → perfect model.

AUC = 0.5 → random guessing (coin flip).

Both models are at 0.997+ → exceptional discrimination.

Live Credit Risk Predictor

The web app lets you enter any borrower's data and get an instant risk assessment from the trained model.



Ideal Borrower

FICO 790 · Income \$120K
DTI 8% · RevUtil 12% · Grade A

Will Repay



Average Borrower

FICO 700 · Income \$65K
DTI 18% · RevUtil 45% · Grade C

Will Repay



Risky Borrower

FICO 620 · Income \$42K
DTI 32% · RevUtil 78% · Grade E

Borderline



Very High Risk

FICO 560 · Income \$28K
DTI 40% · RevUtil 95% · Grade G

Default Risk

The live predictor lets you:

- ▶ Toggle between Logistic Regression and LDA with one click
- ▶ Enter any custom loan values across all 15 origination features
- ▶ See probability bars, risk tier, and top 3 risk drivers instantly

Key Limitation: Data Leakage

The full-dataset models (97–99%) include variables only known AFTER the loan resolves.

- **total_rec_prncp**

Total principal recovered — only exists after the loan ends

- **recoveries**

Money recovered post-default — populated only IF there was a default

- **collection_recovery_fee**

Fee charged to recover defaulted funds — post-event only

- **last_pymnt_amnt**

Last payment amount — known only after payment history unfolds

★ The Analogy

Trying to predict if a student will fail — and using their final exam score as a predictor. Of course the model is perfect. You're using the answer to predict the answer.

✓ Our Live Predictor Fixes This

The interactive predictor uses ONLY 15 origination-time features — variables known at application time. That is why accuracy is ~71%, not 99%. Honest.

🕒 What Real Banks Do

Only application-time data · Must comply with ECOA · Models are audited for discriminatory patterns.

Summary & Takeaways

Thank You

292K

Clean Records

6

Latent Factors

56.5%

Variance Explained

0.997

LDA AUC

0.9997

LR AUC

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Real Data, Real Cleaning

500K messy records → 292K clean rows through a rigorous 7-step pipeline.



Factor Analysis

6 latent factors explain 56.5% of variance in loan characteristics.



LDA

Separated good from default borrowers: 97.3% accuracy, AUC = 0.997.



Logistic Regression

Outperformed LDA: 99.6% accuracy, AUC = 0.9997. Better fit for skewed data.



Data Leakage

High full-model accuracy partly due to post-outcome variables. Honest model ≈71%.



Live Predictor

Interactive web app: enter loan values and watch the trained model decide.